

Elephant Movement in Kaudulla: What the Data Tells Us

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What This Project Is

The Department already collars and tracks elephants in Kaudulla. **This project turns that existing GPS data into patterns the raw records do not show.**

What we did with the data

1. Monitoring missing data
2. Analysis of elephants' movements
3. Climate and vegetation data analysis
4. Checking elephant movement anomalies

The Data We Worked With

#	Dataset	Time resolution	Period	Size
1	Elephant GPS tracking	Hourly	Jul 2024 – Jun 2026	16,747 records, 15 elephants
2	Climate	Monthly	2017 – 2025	9 variables $\times$ 9 years
3	NDVI vegetation classification	Annually	2018 – 2026	5 density classes

Variables Used from Each Dataset

1. GPS Tracking

Hourly

collar_id	Collar number
name	Elephant name
sex	Male / Female
date	Date of fix
time	Time of fix
lat	Latitude
lon	Longitude

2. Climate

Monthly

Rainfall

PRECTOTCORR

Precipitation Corrected (mm)

Temperature

T2M

Temperature at 2 Meters (°C)

3. NDVI Vegetation

Annually

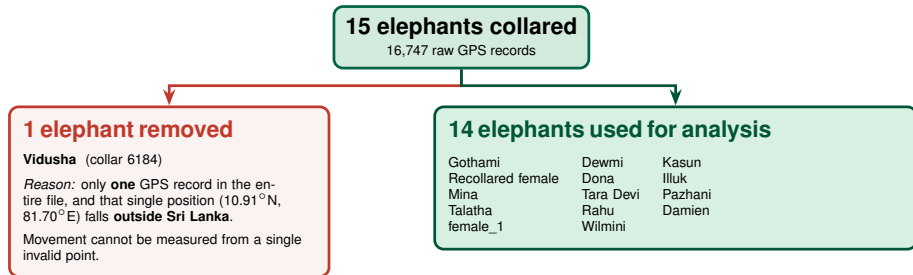
% cover by density class

- Non-vegetation
- Shrubs & degraded grasslands
- Sparse or stressed vegetation
- Moderate canopy coverage
- High density forest

% cover by plant health

- Very healthy plant
- Moderate healthy plant
- Unhealthy plant

Dataset 1: GPS Tracking Data



After cleaning: **16,735 valid GPS records** across **14 elephants**.

Cleaning Criteria Applied to the Tracking Data

#	Criterion	Rule applied	Why	Removed
1	Name correction	Collar 6165 "... " renamed female_1	The unnamed animal needed a stable label	—
2	Valid timestamp	Date and time must be readable	A corrupt time cannot be placed on the timeline	0
3	Coordinates present	Latitude and longitude not blank	No position means no movement	0
4	Inside Sri Lanka	Lat 5.9–9.9, Lon 79.5–81.9	Impossible positions from poor satellite geometry	4
5	GPS accuracy	HDOP ≤ 10	High HDOP means an error of hundreds of metres	8

- **Total removed: 12 records out of 16,747 = 0.07%**
- **No interpolation applied.** Missing hours were left blank, not filled with estimated positions.

Data Availability by Elephant

Collar	Name	Sex	Start	End	Days	Fixes	Expected	% avail.
6119	Pazhani	F	2025-09-07	2025-09-09	2.0	50	50	100.0
6182	Damien	M	2025-09-08	2025-09-09	0.6	15	15	100.0
6156	Wilmini	F	2024-10-16	2024-10-22	6.5	131	157	83.4
6123	Rahu	M	2025-10-20	2025-11-13	24.0	357	577	61.9
6155	Gothami	F	2024-08-23	2025-11-11	444.9	5,373	10,678	50.3
6129	Kasun	M	2025-07-13	2025-07-30	18.0	193	432	44.7
6121	Recollared female	F	2024-08-25	2026-06-17	660.9	5,127	15,862	32.3
6165	female_1	F	2024-08-23	2025-07-23	334.0	2,354	8,018	29.4
6144	Illuk	M	2025-06-22	2025-07-20	27.9	87	671	13.0
6139	Mina	F	2024-07-14	2025-10-27	470.0	1,045	11,280	9.3
6114	Talatha	F	2024-07-18	2025-10-27	466.2	853	11,191	7.6
6131	Dona	F	2024-09-09	2025-08-06	331.0	464	7,946	5.8
6154	Dewmi	F	2024-11-09	2025-11-28	384.0	451	9,216	4.9
6179	Tara Devi	F	2024-07-22	2025-10-09	443.9	235	10,654	2.2

Landscape-Level Movement

All tracked elephants use a common landscape centered around the **Kaudulla Reservoir**, with many also moving towards the **Ralapanawa Nature Reserve**.

Movement Pattern Observation

- An interesting observation is that **Gothami** and **female_1** show very similar movement patterns during September, October, and November 2024.
- **Dona** mainly stayed in the northern part of the park, while **Recollared_Female** showed a unique pattern with frequent visits to both the railway track area and the upper region.

Home Range & Movements

- Kasun and Rahu demonstrated the widest directional movement, whereas Damian and Iluk followed more focused movement routes, indicating individual differences in space use.
- Female elephants displayed balanced movement in nearly all directions, suggesting widespread use of available habitat with only minor variation in directional frequency.
- The per-elephant summary table provides an overall comparison of tracking duration, movement, home range size, GPS data availability, and movement speed for each elephant, allowing individual movement behavior and space-use patterns to be compared across the study population.

Climate-Driven Spatial Patterns of the Re-collared Female

- Her movement pattern indicates that she tends to stay farther from the reservoir during periods of lower daytime and nighttime temperatures.
- Her locations in December 2024 and December 2025 were geographically very similar, with both occurring away from the reservoir. These observations were recorded under the same rainfall category, indicating consistency in habitat use under similar rainfall conditions in same period of year.

Comparison of Daytime and Nighttime Locations

- Day and night GPS points overlap considerably, indicating stable home ranges rather than different day/night territories.
- Although elephants appear to use areas slightly further into the reservoir during the daytime than at night.

Comparison of Hot and Cool Season Locations

- During the hot season, elephants appeared to move further into the reservoir.
- Some elephants, such as Gothami and Dewmi, occupied more northerly areas during the cool season, indicating that their locations were relatively more southerly during the hot season.
- The Recollared Female had the most balanced hotcool dataset (67%, 33%) among the well-tracked elephants. She showed a distinct seasonal pattern in 2024 and 2025, with her cool-season locations shifting away from the reservoir towards the Trincomalee railway line. However, this pattern was not evident in 2026.
- During the hot season of both 2025 and 2026, Recollared female's GPS locations appeared to be more frequently concentrated near a paddy field.
- **Caution:** Since the dataset was imbalanced across seasons, with most elephants tracked only during either the hot or the cool season, seasonal comparisons are limited. Although a few individuals appear to shift their locations between hot and cool periods, the evidence is insufficient to draw strong conclusions about seasonal movement.

Comparison of Heavy and Low Rainfall Locations

- Most elephants occupied similar areas during both heavy and low rainfall periods.
- Gothami showed a slight northward shift under low rainfall conditions compared with heavy rainfall periods.
- Some individuals were only tracked in one rainfall condition, so comparisons were limited.

Vegetation Dynamics and Elephant Movement Patterns

- Vegetation changed slightly across the three years.
- 2026 had the highest canopy cover, but also the highest proportion of non-vegetated land. This is the key vegetation signal that connects to elephant behavior, but 2026 data is only available for recollared female.
- Across 2024 and 2025, most elephants exhibited stable home ranges. However, the Recollared Female showed a noticeable shift away from the reservoir area in 2026 compared with 2024 and 2025.

Monthly Home Range variation

- The Recollared Female had the most complete monthly data and showed the widest monthly home range (4.27 km) in April 2025, at the beginning of the hot season.
- For the remaining elephants, monthly data were concentrated mainly between June and December. Within this period, several elephants exhibited larger home ranges around October/November.